



AERIAL ROBOTICS PLANNING AND NAVIGATION

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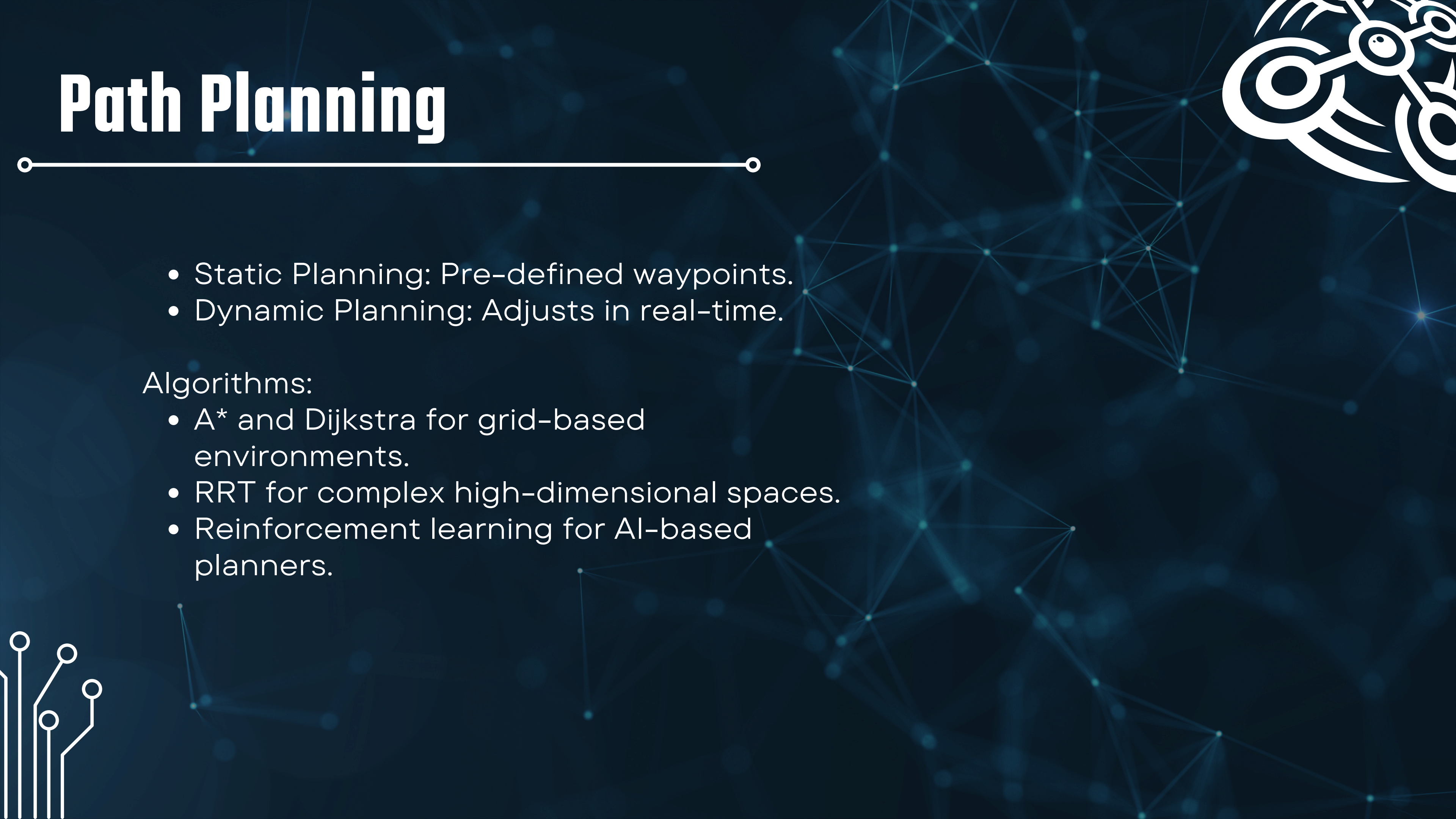
Sense and Avoidance System



- Obstacle Detection: Using LiDAR, radar, vision.
- SLAM (Simultaneous Localization and Mapping): Builds a map while localizing the robot.
- Real-Time Adjustments: Uses AI or fuzzy logic to react to dynamic environments.



Path Planning



- Static Planning: Pre-defined waypoints.
- Dynamic Planning: Adjusts in real-time.

Algorithms:

- A* and Dijkstra for grid-based environments.
- RRT for complex high-dimensional spaces.
- Reinforcement learning for AI-based planners.

Autonomous Control



- High-Level: Decision making (e.g., “go to location X”).
- Mid-Level: Motion planning, route optimization.
- Low-Level: Actuator commands, obstacle response.
- Redundancy and Safety: Critical systems have backups.



Swarming



- Decentralized Systems: No central control; each robot makes local decisions.
- Applications: Search & rescue, area coverage, environmental monitoring.
- Inspired by Nature: Bees, ants, birds; behaviors include flocking, foraging, and aggregation.



Thank You!

